

10.7.51

EE25BTECH11002 - Achat Parth Kalpesh

Question:

Find the coordinates of all the points $\mathbf{P}$ on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, for which the area of the triangle PON is maximum, where $\mathbf{O}$ denotes the origin and $\mathbf{N}$, the foot of the perpendicular from $\mathbf{O}$ to the tangent at $\mathbf{P}$

Solution:

The equation of the ellipse is given as

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (0.1)$$

$$\mathbf{V} = \begin{pmatrix} \frac{1}{a^2} & 0 \\ 0 & \frac{1}{b^2} \end{pmatrix}, \mathbf{u} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, f = -1 \quad (0.2)$$

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} = 1 \quad (0.3)$$

The equation of a tangent to $\mathbf{P}$ is

$$(\mathbf{V}\mathbf{P} + \mathbf{u})^\top \mathbf{x} + \mathbf{u}^\top \mathbf{P} + f = 0 \quad (0.4)$$

$$(\mathbf{V}\mathbf{P})^\top \mathbf{x} + f = 0 \quad (0.5)$$

As $\mathbf{N}$ is the foot of the perpendicular from $\mathbf{O}$ to the tangent at $\mathbf{P}$

$$\mathbf{N} = \mathbf{0} - \frac{(\mathbf{V}\mathbf{P})^\top \mathbf{0} + f}{\|\mathbf{V}\mathbf{P}\|} \mathbf{V}\mathbf{P} \quad (0.6)$$

$$\mathbf{N} = \frac{\mathbf{V}\mathbf{P}}{\|\mathbf{V}\mathbf{P}\|} \quad (0.7)$$

The cross product of two 2D vectors $\mathbf{x}$ and $\mathbf{y}$ can be written in matrix form using the skew-symmetric matrix

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (0.8)$$

Hence,

$$\mathbf{x} \times \mathbf{y} = \mathbf{x}^\top \mathbf{J} \mathbf{y}. \quad (0.9)$$

The area of the triangle $\triangle PON$ is given by

$$\text{Area}(PON) = \frac{1}{2} \frac{|\mathbf{P}^\top \mathbf{J} \mathbf{V} \mathbf{P}|}{\|\mathbf{V}\mathbf{P}\|^2}, \quad (0.10)$$

Let

$$\mathbf{Q} = \mathbf{V}\mathbf{P} \quad (0.11)$$

$$\mathbf{P} = \mathbf{V}^{-1} \mathbf{Q}. \quad (0.12)$$

Then, the area becomes

$$\text{Area}(PON) = \frac{1}{2} \frac{|(\mathbf{V}^{-1}\mathbf{Q})^\top \mathbf{J}\mathbf{Q}|}{\|\mathbf{Q}\|^2} \quad (0.13)$$

$$= \frac{1}{2} \frac{|\mathbf{Q}^\top (\mathbf{V}^{-1}\mathbf{J}) \mathbf{Q}|}{\|\mathbf{Q}\|^2}. \quad (0.14)$$

Since $\mathbf{Q}^\top \mathbf{M} \mathbf{Q}$ depends only on the symmetric part of $\mathbf{M}$, define

$$\mathbf{S} = \frac{\mathbf{V}^{-1}\mathbf{J} + (\mathbf{V}^{-1}\mathbf{J})^\top}{2}. \quad (0.15)$$

Hence,

$$\text{Area}(\mathbf{Q}) = \frac{1}{2} \frac{|\mathbf{Q}^\top \mathbf{S} \mathbf{Q}|}{\|\mathbf{Q}\|^2}. \quad (0.16)$$

Now,

$$\mathbf{V}^{-1} = \begin{pmatrix} a^2 & 0 \\ 0 & b^2 \end{pmatrix}, \quad \mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (0.17)$$

Thus,

$$\mathbf{V}^{-1}\mathbf{J} = \begin{pmatrix} 0 & a^2 \\ -b^2 & 0 \end{pmatrix}, \quad (\mathbf{V}^{-1}\mathbf{J})^\top = \begin{pmatrix} 0 & -b^2 \\ a^2 & 0 \end{pmatrix}. \quad (0.18)$$

So,

$$\mathbf{S} = \frac{1}{2} (\mathbf{V}^{-1}\mathbf{J} + (\mathbf{V}^{-1}\mathbf{J})^\top) \quad (0.19)$$

$$= \frac{1}{2} \begin{pmatrix} 0 & a^2 - b^2 \\ a^2 - b^2 & 0 \end{pmatrix} \quad (0.20)$$

$$= \frac{a^2 - b^2}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (0.21)$$

The eigenvalues and eigenvectors of $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ are

$$\lambda_1 = 1, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \lambda_2 = -1, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (0.22)$$

Hence, the eigenvalues of $\mathbf{S}$ are $\pm \frac{a^2 - b^2}{2}$, and the maximum absolute eigenvalue is

$$\lambda_{\max} = \frac{|a^2 - b^2|}{2}. \quad (0.23)$$

The maximum area is therefore

$$\text{Area}_{\max} = \frac{|a^2 - b^2|}{4} \quad (0.24)$$

The corresponding eigenvector is

$$\mathbf{Q} \propto \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (0.25)$$

Hence,

$$\mathbf{P} = \mathbf{V}^{-1} \mathbf{Q} \propto \begin{pmatrix} a^2 \\ b^2 \end{pmatrix}. \quad (0.26)$$

Imposing the ellipse condition $\mathbf{P}^\top \mathbf{V} \mathbf{P} = 1$,

$$1 = \lambda^2 (a^2 + b^2) \implies \lambda = \pm \frac{1}{\sqrt{a^2 + b^2}}. \quad (0.27)$$

Thus, the required points are

$$\mathbf{P} = \pm \frac{1}{\sqrt{a^2 + b^2}} \begin{pmatrix} a^2 \\ b^2 \end{pmatrix} \quad (0.28)$$

and the maximum area of $\triangle PON$ is

$$\text{Area}_{\max} = \frac{|a^2 - b^2|}{4} \quad (0.29)$$